

Session I Deep Inference with Karl J Friston

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Thank you Amos. Hello everybody. Amos that was a lovely tribute to Mike and and so I won't say too much more other than to say that I can't believe the firefighter problem has come back into our workshop I think so many of us remember the spirited debate that ensued when that firefighter problem first was introduced and so much so that I think that we actually had to declare a moratorium on discussions in subsequent meetings because it kind of took over our debate so it was lovely to to be reminded of that happier time and that happier moment I just want to say that you know Amos mentioned Mike was a constant in the activities of our institute and just for me personally having him as a co-chair on the advisory board made decision making easier and and it was just a comfort to know we were pursuing a shared vision and so I too will miss him greatly and and there's a large hole there that I'm not sure how we're gonna fill moving on to the first session and I don't I haven't seen whether Carl is here but I assume Karl is here somewhere and and I can't tell from the zoom but but let's assume that Karl is here our first speaker I'm delighted to introduce is for our invited session is Karl Friston he's a theoretical neuroscientist and an authority on brain imaging he invented statistical parametric mapping voxel-based morphometry and dynamic causal modeling these contributions were motivated by schizophrenia research and theoretical studies of value learning formulated as the disconnection hypothesis of schizophrenia his mathematical contributions include variational laplacian procedures and generalized filtering for hierarchical bayesian model inversion he currently works on models of functional integration in the human brain and the principles that underlie neuronal interactions his main contributions to theoretical neurobiology is a free energy principle for action and perception known as active inference his awards are way too numerous to list here but one thing I will note I asked him whether there was anything he wanted me to add and he said I was free to do that and and some of you know how dangerous that can be but I will add that um that I noticed that he's got and and this is as an economist this is just massive he's got over 144 google scholars like 144,000 google scholar citations so with

without further ado and it gives me great pleasure to introduce Karl Fristen Karl take it away thank you very much that was one indeed one of my favorite facts about me that you that you picked up let me just get myself organized and then well again thank you for that lovely introduction and it's a great pleasure and honor to to be invited to speak to you and particularly on this sort of sad occasion I'm intrigued by the firefighter problem that's sounds something I'd like to discuss even if you're not allowed to discuss it I'm not sure I'm going to get to that depth or problem or challenge my hope was in the next 30 35 minutes was to introduce a perspective on evidence accumulation the handling of information from the point of view of normative models of sentient creatures, creatures like us or indeed viruses and plants and I'm going to take my lead from a simple assumption that every self-organizing system every information processing system is at the end of the day thinking and behaving in a way to maximize the evidence for its models of the world and I'm going to try and motivate that from a sort of machine learning perspective practically and as the talk goes on show how these ideas translate in terms of the sentient behavior of biotic systems like like animals including ourselves as has been mentioned this comes under the rubric in the life sciences of as active inference active sensing sensing in the spirit of trying to disclose those data of that sensory information which resolves the greatest amount of uncertainty about your model or hypothesis about how those data were caused now if we have time i'll give you a brief simulation of the sort of epistemic foraging that ensues from this so self-evidencing is very simple all it's saying is you can account for most perspectives on behavior in terms of this single underlying imperative which is that your internal states of the processing system say internal brain states and the way that you act upon the world the things that you can control on the outside are both there in order to maximize the probability of some observables or outcomes given or conditioned upon me or a model of how those outcomes were generated and I'm just motivating the simplicity of this assumption from the perspective of a number of global approaches to behavior and self-organization in the life and sometimes the physical sciences so if these outcomes are the most likely kind of outcomes that I would expect to encounter then they define operationally my characteristic outcomes to which I will aspire and apparently work towards so in economics you could think of the log of the probability of an outcome given it is experienced or witnessed by a particular model or me as standing in for a value or a utility and from that one can spin off reinforcement learning optimal control theory and indeed in economics expected utility theory that's interesting because the negative quantity of this way of reading value is self-

information surprising or more simply surprised that means that this imperative here the self-evidencing simply translates as the imperatives to maximize the mutual information for example the infamous principle in neuroscience or the principles of minimum redundancy maximum efficiency from Horus Barlow and indeed the free energy principle which we have been involved with where this free energy serves as a bound on this quantity and self-information here that itself in turn is interesting because the time average of self-information is entropy which means that this imperative is self-evidencing and can also be read as the holy grail of self-organization many to minimize the dispersion or the the entropy or the dissipation of my observable states that you find in theories of self-organization ranging from cybernetics synergetics and of course if you're a physiologist this is just a statement of homeostasis keeping physiological states within viable bounds but I'm going to take another perspective on this quantity and treat this quantity as the probability of some data given a generative model where in basic statistics that would be called the the bayesian model evidence or more generally the marginal likelihood marginalized marginalizing over all the parameters that engendered or generated those data there and in my world that leads to things like the bayesian brain hypothesis formulations of evidence accumulation predictive coding in engineering and now indeed in the neurosciences so I'm just going to now unpack that kind of self-evidencing in a slightly more technical way with a special emphasis on this variational free energy that serves in machine learning as a bound on the log evidence in the sense that we can create a variational bound simply by equipping the log evidence with a quantity that can never be less than zero namely a k-l divergence and in this instance it's a k-l divergence between the some approximate or bayesian beliefs about states of the world generating the data and the true posterior over those states if I were able given those data so by optimizing extremizing this free energy and we have on the one hand a convergence of our beliefs encoded by our internal states to the true posterior whilst on the other hand this bound now becomes a bound approximation to the log evidence that this self-evidencing principle aspires to maximize and just to motivate why this might be a useful perspective and what you get from this is essentially explainable artificial intelligence in the sense that having a generative model there commits you to writing down a way in which your data were generated that you can explain to somebody else it also requires you to make explicit and transparent your prior assumptions that are entailed by the form and the structure of that generative model and any hyper parameters associated with it in a sense it also gives you a design optimality in terms of making optimal decisions of an objective sort in terms of and

using this model in order to make inferences about the causes or the way that your data were generated and that itself we will see leads to a principled way for data mining or data foraging and indeed epistemics and generalized artificial general intelligence particularly with sparse data we'll come to that but there is another perspective though or another way of carving up this variational free energy functional and so just by rearranging the terms I can also read it as a mixture of accuracy and complexity so essentially the free energy is accuracy minus the complexity where the complexity here scores the difference between my bayesian beliefs after seeing some data and my prior beliefs so literally the degrees of freedom or the amount to which I have changed my mind in the face of some data and that complexity term plays an important role in many guises is it's just an expression of Occam's Principle in the brain it leads to things like factorizations and functional specializations a simpler way of explaining or making sense of data with minimum redundancy and maximum efficiency if you're an engineer it also implies that you're aspiring to provide an accurate account of observable data with the minimum computational cost leading to notions of bounded rationality literally based upon an evidence bound known in my world as approximate bayesian inference that rests upon priors that sometimes people refer to as heuristics and of course one can appeal now to the physics of information by the Drazinski equality and Landauer's principle which means that minimizing computational cost also implies a thermodynamic efficiency that in principle the best way of accounting for data or making sense of data should be that way that uses up the least energy and is performed in the minimum amount of time so the basic story I want to tell in this short presentation rests upon thinking about well that's fine for understanding sentient behavior and a principle for sort of optimal action in the moment what would it mean if I considered devices artifacts or creatures that can plan into the future and have a generative model of the consequences of their action and what would this self-evidence imperative mean for plans and the kinds of behavior we'd expect to see under that imperative and the answer is or one answer at least is very simple your because now I am trying to maximize or optimize a free energy bound on evidence but in the absence of actual outcomes the outcomes become a random variable and I can simply take the expectation of this evidence bound under the predictive distribution of the outcomes that would ensue if I did that and if we follow that through what we're going to see is that basically we can explain the best kinds of plans the self-evidencing plans as fulfilling two imperatives one that of bayesian optimal base optimal design experimental design in terms of maximizing expected accuracy whilst making bayes optimal decisions by

minimizing risk by minimizing the expected complexity and I'm going to basically rehearse the observational suggestion that minimizing expected sorry maximizing expected free energy is basically a mixture of these two things just to make this a bit more heuristic what I would normally do to your psychology audience is ask them to imagine that they are an owl and that you're hungry and then I'd normally ask a member of the audience what are you going to do I won't do that now but I'm hoping that you're all thinking well I'm going to go look for food indeed that is the absolutely the correct answer but in answering like that what you have said is something quite fundamental and I'm going to try and unpack how important that answer is just by comparing and contrasting two ways of writing down normative or objective functions for good behavior and I'm actually going to repair these two perspectives you know in a few slides but I'm going to deliberately sort of introduce and celebrate this distinction or this dialectic first of all I could assume that if I make a move on the world I implement a particular action you then there are going to be certain states in the world that ensue and I may have a value function over those states and if I had a value function of states then there exists a policy that is defined in terms of the best action to take from any given state namely that which maximizes the value of the ensuing state however there is another way of formulating good behavior which speaks to the answer I'm going to look for food because looking and searching can be construed as a way of minimizing uncertainty in this instance where the prey is and uncertainty is an attribute of a probability distribution I'm going to refer that as a bayesian belief or a conditional probability distribution so that tells us something really important it says the best behaviors the best actions the best plans are cannot be functions of states of the world they have to be functionals of beliefs about states of the world and that answer looking for food also tells you something else because it matters whether I look for my prey and then I eat it or whether I try to eat it and then look for my prey so that introduces a notion of sequential policies or sequential actions that have to be optimized in relation to this functional of um of beliefs about states of the world and I'm going to frame that that aspect of optimizing a trajectory or a sequence of plans in terms of a principle of least action where action here is just the path of the time integral of a function of beliefs and indeed it's going to be the expected free energy and hence the path of the time integral of the free energy is an action so what we have at hand here are two very different ways of thinking about sort of a good way to exchange with our world on the one hand you've got bellman's optimality principle that rests upon a value function of states and many of you will be familiar with that in terms of optimal control

theory dynamic programming deep reinforcement learning basic decision theory and so on but there is another way of thinking about it and that's possibly slightly simpler appeals to hamilton's principle of stationary action or at least action and that's really um the principle that the free energy principle inherits from we'll also see that this is very closely aligned with notions of artificial curiosity and intrinsic motivation in robotics the principles behind acquiring the right kind of data optimal bayesian design and sequential policy optimization so I'm just going to take you through the functional form of this objective function one more time because I think you know its simplicity is compelling and it speaks exactly to a variational or an appeal to hamilton's principle at least action as possibly another way of thinking about behavior or at least sentient behavior so this is the the basic setup that we use to understand and simulate and indeed model behavior and choices that are informed by observations we have some observations that come in they are used to optimize our bayesian beliefs about hidden states of the world behind the observations generating those observations and we do that by finding the most accurate account that's as simple as possible of those data and then we take those beliefs about hidden states or latent states of the world and then we use them to evaluate the expected free energy given a particular policy and in so doing we're effectively taking the expected complexity and the expected accuracy conditioned upon a way of moving forward and that equips every way of moving forward every policy with a score and expected free energy that can be associated with a long probability of me doing that and then I can just apply a soft max operator find the most likely plan select my next action change the outside world that will generate new data and so the cycle continues so that's the basic idea is exactly the same equations that I've unpacked as previously just to reiterate that we can either regard this free energy function on this evidence-bound excuse me as a mixture of complexity and accuracy or we can just rearrange it to express it as a bound on log evidence as you do say in machine learning or deep learning with things like variation auto encoders and the reason I've done it like that is just to see what happens to these terms when we take the expectation to form the expected free energy and what happens is the complexity becomes a risk the inaccuracy becomes ambiguity the evidence bound becomes something uh known in um economics and robotics as intrinsic value or motivation and the log evidence becomes an extrinsic value that we'll see in a moment can be associated with things like expected utility so what licenses my interpretation of these expected terms well let me just go through and unpack this expected free energy by looking at a number of different special cases let's pretend that my prior

beliefs about outcomes given I am me encode my preferences my prior preferences about the sort of outcomes I expect to encounter following all actions and let's assume I'm agnostic about what happens I have no particular preferences for this or that outcome that just leaves this quantity here which is the expected evidence bound and this quantity is just the mutual information or expected mutual information between the causes and the consequences under a particular plan of soliciting those outcomes there in neurobiology particularly in the visual search literature where you people try to work out where am I going to look next you know what motivates that visual palpation of the world the motivation is just the intrinsic value of resolving uncertainty called bayesian surprise more I think revealingly is just the degree of belief updating or information gain afforded some outcomes in other words the difference not the k-l divergence between my bayesian beliefs about hidden states if I were able to see the outcomes that ensued following a policy compared to my beliefs before looking at those outcomes again speaking to the difference between a posterior and a prior but now in the future so it's an expected divergence or belief updating or information gain that is intrinsically valuable irrespective of what you think or will prefer to happen let me now take another source of uncertainty off the table namely the ambiguity I'm going to make a simplifying assumption that I can see all the hidden states of the world that my sensations my sensory observables are a direct mapping of the states of the world that generated them and in this instance the s's become the o's and we're only left this ambiguity term disappears and we're just left with this quantity here which is another k-l divergence or relative entropy here namely the difference between my anticipated outcomes or states given a particular policy and my preferred outcomes my prior preferences so all we're saying here is that the actions that have the optimal expected free energy are just those in the absence of any ambiguity that minimize risk as scored by the difference between what I anticipate will happen and what a priori I prefer to happen and if I make a final move which is to take the last kind of uncertainty out of the game namely the uncertainty that depends about states the world that depends upon my action that I'm just left with this term here the extrinsic value but if you remember we're associating with utility in economics or negative loss functions in optimal control theory which just means we're trying to optimize the expected long probability of our preferred outcomes or expected value here so what we have at hand is a decomposition that applies to policies when scored in terms of an expected free energy functional and that can be unpacked or carved into an expected value that underwrites these optimal decisions but importantly this epistemic term this information gain that is informally

identical to the principles that underlie bayes optional design and things like active learning with unknown parameters so in short the expected value plus the information gain is literally equal to the expected free energy I slipped this slide in because I thought some of you might like to think of it in terms of information diagrams but given the time constraints um and the opportunity for discussion I'm not going to go into that it's just an interesting game that you can play in terms of thinking about an information theoretic partition in terms of um basing design information bottlenecks empowerment another takes on the right kinds of information gathering behavior I want to whip through now a simulation because you know we want to leave as much time for any discussion that sort of exemplifies how these mechanics would unfold in empirical behavior in my world here looking at sort of small animals or rats making decisions what we normally do is build a generator model based upon a markov decision process of discrete states that generate outcomes parameterized by a likelihood of mapping a and then we make the transitions over time among the hidden states depend upon some policy depending upon property transition matrices where the policy itself rests upon my prior preferences or some cost function and then I can equip this model with a few initial priors and hyper parameters I have now got a completely a fairly complete description of any kind of behavior that can be written down in terms of discrete updates of hidden or latent states that generate observables I can then apply standard variational techniques to optimize my variational and expected free energies essentially using off the shelf technology that rests upon defining this approximate posterior just from really from my interest when one takes these off the shelf belief updating basic belief updating variational message passing or belief propagation schemes they look remarkably similar to the way that neuroscientists understand message passing and belief updating the in the brain in terms of perceptions updating beliefs about expected states of the world policy selection usually associated with the striatum or integration between parts of the surface of the brain and the striatum that rests upon this expected free energy and indeed some hyperplasia or hyper precision parameters reflecting the confidence or the uncertainty that I afford my beliefs about what I'm going to do next we can do learning and then we just select the action on the basis of these updated expectations about policies so when we do that and I'll just close with an example of a synthetic rat doing this so this little paradigm here involves a rat in a tea maze it can go and secure a reward that's either on the right or the left of the two upper arms it could only make two moves but it's got the opportunity to go to the lower arm and get an instructional cue that tells it that the cue is the the war reward is on on the left or on the right

hand side so it's got a choice it can either take a gamble and go to the one of the two upper arms and be right or wrong and it has to stay there so these are absorbing states when it gets to the baited areas or it can waste one of its moves by finding out exactly where the reward is and then go and secure its reward with a high degree of uncertainty so from the point of view of expected utility these are fairly balanced options but clearly from the point of view of this self-evidencing in the future the information gain the intrinsic value of going to resolve your uncertainty or this rat's uncertainty about the states of affairs is going to be more attractive and that's going to compel the rat to be exploratory and go and look at the cue get that and then go and get its reward and indeed that is what it does i'm just summarizing the behavior that follows from integrating those belief propagation or variational message passing schemes under this charity model but with a twist after the first couple of trials we leave the reward here so that in principle the rat is going to start to learn that the reward is always here and indeed it does that and of course as it becomes more and more confident that the reward is here the intrinsic value the epistemic value the information gain afforded by going down here gets smaller and smaller and smaller and wanes and once it passes a threshold where the expected utility of going straight to the reward and staying there for an extra move and supervenes then its behavior switches deterministically and it switches from an exploration to an exploitation mode and that's exactly what happens in this example after about the 20th trial here and all this depends upon accumulating knowledge about the contingencies namely the context where on which side is the reward most likely to be over successive exposures that then couples back into this online active influence planning as inference underwritten by this expected free energy this is the last slide I'm putting in this in just to remind myself that there's an outstanding challenge here I've sort of um sold this self-evidencing as a something that has constructed into in relation to lots of other ways of describing good behavior there may be a deeper backstory to this that relates to statistical physics and things like the integral fluctuation theorem but to connect or complete that story it requires one small move which I'm happy to talk about um but not happy to waste any more time unpacking for you at the moment so i'll give the last word to Einstein that everything should be made as simple as possible but not simpler so it just reminds me for me to thank those people whose ideas I've been talking about and of course thank you for your attention thank you very much indeed terrific Karl thank you thank you so much for that really engaging talk and a great way to start off our workshop I realized I actually forgot to mention to everybody else who's here that that what the format was going to be so we

did actually plan for some ample time for questions and for discussion I think as we know as people who have been to our workshops before know that's one of the hallmarks of our workshops is a quite animated discussion and so thank you Karl also for leaving ample time for that so I want to open the floor up to discussions and questions maybe if you want to just drop a note into the chat to me just to tell me of a question so that we can organize or line up the questions but I think the first question I know almost has a few questions so we'll go to Amos first but just drop me a note as I said if you have questions to ask thank you Karl first it's nice to see you again I really enjoyed your talk and the idea that you were able in such a simple way to put decision theory within information theoretic framework in quantities mostly the k - l divergence seem to be very interesting very nice I myself even though I work on this area for many years I never really saw it in in this way so thank you for this I have a simple question though it could be outside the scope here in a previous time I've seen you were talking about Markov blanket and as as you were passing saw in one of the slides I saw that you put your framework within a Markov blanket maybe I missed it and it may be outside what you wanted to talk today but if it's not I would be happy to hear a little bit more how this fits within your theory of Markov blanket that's a very astute question yes I tried to slip past the Markov blanket because it would take another half hour to motivate that but that's an excellent question because it speaks to that last slide where it were once trying to find a deeper backstory in statistical physics to why this kind of self-evidence behavior has to be there if any system manages to maintain itself in some kind of steady state you know if it's in computational chemistry this will be self-assembly if it's in biology it be autopoiesis if it's in social neurosciences it's just basically maintaining cohesions and and in groups and and that's where the Markov blanket comes in so in order to talk about a system you'd have to actually write down well what do you mean by a system that has some states how are those states differentiated from the states that do not belong to the system so that implies a set of conditional independences that as you've noted um are usually assigned to the blanket states the Markov blanket so what you say is well in order to differentiate a system from the rest of the universe I will assume the existence of some states that when I condition the internal states on they become conditionally independent of the external states and those are the blanket states and then you look at the the density dynamics that you have under that particular partition of states into inside and outside separated by blanket states that still permit so the system's still open because you can the outside can vicariously influence the inside and vice versa through the blanket states but you then look at

the the statistical physics of that partition and that's where the integral fluctuation theorem kicks in which is why I was trying to and I'm still trying to connect the optimum bayesian design plus optimum bayesian decision theory formulation in terms of information theory and k-l divergences with the the k-l divergences underwrite the integral fluctuation theorems so that's I think I mean it's a little bit indulgent but I you know it really does if you like join the dots between the statistical physics and the more practical aspects of information gathering and exploitation thank you great thank you Amos I think our next question is from Bill Lawless I know Bill you put it in the chat but go ahead and unmute yourself and just ask it thanks for reminding me to unmute I it was really nice talk I liked it a lot it seems to me to be based on the information that's available or that could become available and that leads me to my question it seems that the only way that you can uncover something like deception is when it uncovers itself maybe you could address that but I'd also like to find the link between which I like bays and maximum entropy production maybe you could also address that as well if you have time yes we don't have time to address it properly but we can certainly celebrate the existence of those those important issues and questions and the yeah the maximum entropy principle both as it pertains in you know in the sort of classical jamesian sense but also as it pertains to entropy production in this sort of dynamic self-assembly formulation in terms of the integral fluctuation theorems and markov blankets plays an incredibly important role and just you know almost emerges from for free from the free energy formulas so if you just decompose the complexity you've basically got the entropy of your beliefs about the causes of the data and that has to be maximized as mandated by things like Occam's principle so that's part of this sort of optimum-basing design formulation and you know has its roots in sort of jeffrey's priors and james's interpretation of fiscal physics and that so that is really at the heart of this um belief based or at least conditional probability distribution or bayesian belief based formulation of optimal behavior the deception issue is fascinating because it speaks to the rather delicate and challenging and very real issue of applying this mechanics to dyadic or multi-agent games where now you have the potential of the generative model that sort of in this instance that markov decision process being a model of another model so somebody that I am in exchange with in a game theoretical in a sort of language sense and just you know the very notion of trust and deception or indeed just the very notion of a generative model being able to explain outcomes generated by another generative model you know is deeply challenging and we're talking about you know sort of metacognition through to theory of mind here so I think just

things like deception and regret are if you like the you know the most challenging kinds of information exchange or the most challenging aspects of exchange data exchange that you would aspire to try and explain in terms of your bayesian optimality principles in this framework I'm not sure it's been done yet there have been baby steps in terms of putting these sort of free energy minimizing agents together and what tends to happen is that they tend to cooperate because they want to make themselves mutually predictable so if you're trying to minimize the surprisal or the self-information one way of doing that is just to make sure that I am as much like you as possible under the prior assumption that you are doing the same and then our entire exchange becomes as mutually as predictable as possible and that and that optimizes then the joint free energy that that we share did you have anything deeper in terms of deception because my answer is a little bit superficial in the sense that it hasn't quite been done yet yes I do but you mentioned the word assembly which I find to be even more intriguing one of the issues that I have with von neumann's assembly idea is that there's no way to tell when the correct pieces from von neumann's model I don't know if that's what you're addressing there's no way to know that the assembly is working in the right direction that you're actually improving the situation with assembly maybe you could address that not in a scholarly way because I don't know the the you know the notion of assembly in the von neumann sense so that's I'll have to put on the list with the firefighter problem just to go google afterwards simply in any sense how do you know that you're assembling a team based on what you've said you want to assemble it so that the surprise is minimized so one way of addressing that is well certainly through numerical analyses and that's been done in the context of dale's interactions but also an interesting dance between the environment and a phenotype or an assembly or ensemble of phenotypes in the context of economic construction and the bottom line is that the because the this variation free energy is an extensive quantity you can read the goodness of the assembly or the ensemble of systems as simply the sum of each individual's free energy so if every individual of an assembled ensemble is optimizing its free energy then collectively that is also true and then that sort of brings us back to what I was saying before that you know that very much means that everybody's trying to learn about each other so that they can render everything else or indeed the environment as predictable as possible so we build traffic lights and roads and we have languages and signs and deontic cues or I learn your language while you learn mine so that that's sort of the first order emergent property of this kind of multi-agent assembly if that's what you meant but I do note of course that doesn't leave any room for deception so

now you have to think about in groups and out groups assemblies of this kind versus that kind and how you might actually use deception to exploit this sort of first order cooperative mutual predictability imperative that comes out of self-evidencing with other things like me I have a somewhat maybe it's related question but so as as you were presenting I was jotting down a note I made to myself and this is related to things that I'm interested in about the relationship so as an economist I'm quite interested in people's perceptions and how sometimes you know when we observe behavior that we would often characterize as not rational it actually is probably behavior that is rational conditional on a set of beliefs that are possibly not correct or perception that's possibly not correct and so I was wondering about the distinction between perception and what I would call something like learned experience and your slide on functional anatomy and message passing seemed to discuss that and you seem to emphasize that beliefs are about uncertainty and so I was wondering if there was anything that incorporates uncertainty associated with the learning side so there's some uncertainty related to the outcome the results from repeatedly doing the same thing so you know and I think of in a rat experiment you can design it so that there's a constancy of the experience so that after 20 times the rat knows where the food's going to be but and presumably you can design a rat experiment where there's some uncertainty as to whether the food will be there but I was wondering about that about how experience how you think about learned experience and and how an individual knows whether it's their perception or whether it's actually a series of of experiences or you know basically there's there's a dependence in there that I wasn't clear how to incorporate into your models right no excellent question and that dependence is a really important part of approximate bayesian inference that rests upon a mean field approximation or a factorization of beliefs about different kinds of things that when you actually do the belief updating necessarily it you know repairs that dependency so that depends is absolutely crucial and central to any realization of this obviously this amazing belief updating and in particular beliefs about sort of time invariant parameters of a generative model that encode laws and contingencies versus quickly changing states in the moment so normally we talk about inferences basically belief updating about latent states of the world while learning is a slower updating of the parameters of a generative model but you you bring up so many intriguing issues and I just want to mention a couple of bounded rationalities seems to me semantically and possibly rather cheekily easily absorbed into the use of an evidence bound to move from exact basic inference to bounded bayesian inference which sounds to me exactly like sort of bounded

rational and you could even chase up further with the complete class theorems though as you say for any given pair of loss functions and behaviors there is a set of prior beliefs that render that behavior renders that behavior Bayes optimal so there's a lovely duality there that all rests upon the prize that define the generative model so that's that I think that's a really important point that you know we're all different because we're different priors but we're all optimal well at least give them those priors practically you also brought the so the inference in learning are both in the service of minimizing this path integral of free energy that in an identical way that when you unpack the equations analytically you get the sort of fast belief updating that our neurons might do for the inference part then you get this slow accumulation of evidence and belief updating on connectivity that underwrites the parameters and when you start to do numerical analyses something really simple but I think very telling happens that if you set off your I'm sorry I should say that in practice what we generally do with discrete state space models is parameterize the likelihood mappings and the prior usually transition matrices in terms of dirichlet parameters so you have to write down your initial dirichlet accounts now if you write them down very very small the artifact then becomes very sensitive to evidence so that you know if you start off with your two outcomes and you've got one dirichlet parameter after just one experience one's going to be sort of twice the other on the other hand if you start off with 200 dirichlet accounts and you add one more you've got a ratio of 201 to 200 nothing has really changed so your prior beliefs encode an uncertainty about your beliefs about the contingencies that you're learning so if you come in as a sort of young artifact that's just been born with very low divisional accounts it'll be very sensitive to any evidence that is accumulated through repeated exposure but if you're an old wise recalcitrant rat you're not going to learn anything and then it's interesting when you come back to the previous discussion you know when you put two of these agents together they're trying to predict each other if you put the old one with lots of dirichlet accounts and interacting with the baby one with very small dirichlet accounts then the small one learns a lot more from the old one so it's a nice metaphor for you know for teachers and parents and like sorry that's quite interesting I mean I can imagine a situation though where there's I mean in some ways that that you know young one and how it acquires experience is in some ways I think of as an initial condition but over time you can imagine something where there's like you know some limited memory or memory decay so that in some steady state there's you know it's not the marginal additional piece of information has the same effect on all the different agents now I don't want to take up all the

time though because I know Duncan Foley also has a question so I'm going to hand it over to Duncan Foley as well for his question well thank you for your presentation I'm particularly intrigued by the ability of your ability to bring out the parallels to the thermodynamic concepts in this way of partitioning the free energy relationships that are implicit in the entropy reasoning I wanted to go just get your thinking on a slightly different topic which is what I would call novelty because I'm aware that in some models of learning an important role is played by the ability of the system to create new categories or new states or not just to update in the typical bayesian way but to innovate in some way or to accommodate novelty and often there's a trade-off between the organism's willingness to view something as a novelty and start treating it as something different we're trying to accommodate it within the framework that's already been established by experience I just wondered how what your thinking is on how that relates to this general perspective of bayesian updating learning I think it relates intimately and actually provides a sort of super ordinary context and interestingly actually speaks to Robin's last question which I would sort of frame in terms of the problem with structure learning and when do you build an extra bit of your model to account for this new piece of data or indeed remember there's also this imperative to keep the model as simple as possible so yes you have to have the optimal forgetting you have to have the loss of redundant connections you have to you have to forget stuff so it introduces this really delicate issue when is it optimal to believe update or indeed change the structure say the new number of hierarchical levels in a deep network or add another level to some hidden factor so from the point of view of optimizing the free energy bound on model evidence this just is basic model selection so structure learning just is basic model selection under some prior beliefs about the volatility of the environment and there will be an answer to your question should I add an extra level of say hierarchical depth to my model or not in light of these data so you know if you had all the time in the world then what you would do is you'd basically evaluate the marginal likelihood or the evidence bound with and without that extra level and if the evidence the marginal likelihood increased you retain that extra level and if it decreased then you would you lose the extra level and I frame it like that because there are some people who think that's why we go to sleep that we rehearse all the data that we've accumulated during the day and effectively whilst doing things like dreaming evaluate its capacity to explain those kinds of data and then identify those redundant parts of the model structure and remove those connections and simplify and do housekeeping of a structural sort so again it's speaking to this sort of optimal forgetting that

Robin was alluding to and I suspect you know the thrust of your question which is which is you know when do I update when I don't update that would be one part another really important part of your question there was a notion of novelty and novelty emerges and in this scheme in a very simple way it's simply the information gain or that intrinsic value not about hidden states of the world but about the parameters that we've just been talking about so if in my evaluation of the expected free energy of a particular move say looking over there for example or doing this or opening that door there is certainly an information gain an epistemic value or affordance relating to the resolution of uncertainty about what's behind the door but there's also an epistemic value or intrinsic value in terms of what would happen if I do that generally and that's basically scored by the k-l divergence between the posterior beliefs encoded in by those dirichlet parameters or the likelihood mappings or the priors before and after making making that move it's actually very very simple to evaluate so that you've now got sort of two kinds of information intrinsic value you've got the kind that scores the reduction of uncertainty the information gain attributable to bayesian beliefs over hidden states and the other flavor which we can refer to say salience and the other kind that has mathematically exactly the same form but it's a reduction a relative entropy or a reduction of uncertainty of beliefs about parameters that you can think of as novelty so that hasn't been explored very much but it becomes absolutely crucial when you start to try and explain sort of higher end executive functions in cognitive neuroscience like things like sort of aha moments or insights that require you to go and learn something about the causal structure the regularities that endure over multiple instances of a particular trial or a particular self-sensory experience so in my world in the active inference world that is starting to attract a lot of young people in terms of how they might simulate and write that down thank you great now I noticed Kendrick Nelson posted uh something in the chat Kendrick do you want to do you want to mention it or is it fine that it's just in the chat it's not really a question I can just explain it a little bit and then if there's interest people can follow up but you know as people are asking questions about how you model deception and degrees of belief in updating data one of the things that came to mind is there's some really interesting work by Fleming Topsoi who looks at these information theoretic models and games game theory and complexity and he uses generalized entropies to model your willingness to update your beliefs and particularly you can parameterize it where the control parameter becomes a your kind of perception of risk or your perception of complexity and then you can have different players react differently to you know information games where

one player might you know just accept reality and thereby you know when they get new data they just accept that data and update whereas another player might be skeptical about reality and and be less willing to change their beliefs so it's a way of modifying these models to account for biases and how people accept or reject data and I think very important if you take these ideas into the clinical realm in terms of psych psychopathology and psychiatry you know they're thinking about things like delusions and hallucinations or agnosias where people have very different interpretations about what's causing their sensations and afford very different weights to the sense of evidence at hand versus their prior beliefs so I think that's a really important perspective just technically I don't know this formulation but it sounds as if you would reproduce that that mechanics by hyper parameterizing the precision of various beliefs so you know if you want to make a rat more exploitative as opposed to explorative you can certainly increase the precision of its prior preferences so that it's the kind of rat that very definitively likes these kinds of outcomes as opposed to those kinds of outcomes and there's a similar way you can you can parameterize the precision of likelihood mapping so if a likelihood mapping is very imprecise you're going to ignore a lot of data because you know that the predictability of that data or its decision is low but on the other hand if you assign your likelihood mappings undue precision you will be enslaved and your belief updating will be very susceptible to any data and you may well start to overfit that data in relation to your prior beliefs so I don't know but it sounds as if that's the set of questions about you know the relative weighting of different sources of information and the advantage of waiting of sensory likelihood you know versions of priors that are being addressed I'll just reiterate that sounds that sounds important because it's in computational psychiatry that is usually the seat of pathology that explains a lot of false inference and updating and in psychiatric conditions yeah yeah right so everybody I think it's time especially I'm cognizant of the fact that in this virtual world we have to keep on schedule so I'm going to close this here but I just want to first of all I want to thank Karl for for an excellent presentation I want to thank everybody else for the great discussion and and just like in a regular you know sort of real world or in-person conference I mean Bill's already posted a follow-up comment that I think you know hopefully that I find very interesting so hopefully we can take that into the break we can continue this discussion on the break but I just want to close the session now so there's people who think 10 minutes and do whatever they can do that but again thank you so much Karl and and hopefully we can continue to have discussions into the break thanks very much Karl if you're still there I think Bill had an interesting question and that that it

might be worth the having some discussion about right sorry yes I was just about to get a cup of coffee so I'm sorry no coffee for presenters I'm sorry about that yeah this is the quantum biology quantum models of biology wait Bill just Bill just ask your follow-up question yeah it's the owl problem now the owl waits for prey to uncover itself but owls are also built in with the intelligence that prey hide so it's got to go into the business of uncovering a deception volkswagen had an extraordinary deception how do we uncover these deceptions ahead of time that's the question I see that reminds me the stag hunt paradigm have you do this is cooperation versus competition in terms of yeah and you know I don't have an informed answer other than to say that you know this is sort of high-end generative modeling in either a game theoretic or game theory sense or in a social neuroscience sense and the trick is to build the sophistication of the generative models so that they can entertain or represent the possibility that the other agent knows about the you know their intentions and that requires quite sophisticated quite sophisticated generative models and then one gets I presume into the same realm of sophistication that is found in economics you know you to what extent do I represent your representations of me and then to what extent do I represent your representations of you representing me representing you yeah I want level of recursion does you know do you find that sort of that optimum in terms of this sort of bounce between the accuracy and the complexity I don't know of any formal work within the sort um the active inference scheme but I do know that there's a chap called john danizo who has looked at evolutionary stable strategies in using sort of levels of recursions of beliefs in terms of dyadic interactions and actually found something quite quite it wasn't quite so much the prey camouflaging or deceiving the predator but it was more about sort of populations of two different kinds finding that evolutionary stable strategy as quantified by a minimum or an optimum expected free energy and what he found was that the evolutionary stable strategy was half the people had to be very unsophisticated and the other half had to be very sophisticated and that was the only stable strategy otherwise you got this sort of leapfrogging where you know the deceivers try to deceive the deceivers deceiving the deceivers but there was a stable strategy where there were basically sophisticated non-sophisticated agents out there I'm not sure it's framed in terms of deception though i think this is more just in terms of communication well thanks for your answer I'm sorry to have interrupted your coffee right I'll go so I've got to do deception fire fighter problems and what was the other thing oh yes the the von neumann assembly these are things I have to go and do my homework on